Wage Inequality in Turkey

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1 Introduction and Definition

The purpose of this study is to understand wage inequality in Turkey. It aims to explain how wages are distributed among workers and why some people earn much less than others. By using recent data, the study looks at the size of wage gaps, who is most affected, and what causes these differences. It also compares Turkey to other countries to show how serious the problem is. The goal is to help people and policymakers see why wage inequality matters and what can be done to reduce it. In short, this study helps us learn more about fairness, opportunity, and economic justice in the Turkish labor market.

Wage inequality means that people earn different amounts of money for their work. Some workers earn a lot, while others earn very little. For example, a factory worker and a manager in the same company can have very different wages. This is a big problem in many countries like Turkey where wages are the main income source for most people. It affects how people live, how the economy grows, and how fair the society is.

Wage inequality is different from income or wealth inequality. Income inequality includes all types of income, such as wages, profits, and government support. Wealth inequality is about who owns houses, land, and savings. But wage inequality only focuses on what people earn from working. This makes it very important to study, especially when we want to understand problems in the labor market.

There are many ways to measure wage inequality. One is the Gini coefficient. A Gini score of 0 means everyone earns the same. A score of 1 means one person earns everything. Other tools include comparing the top and bottom wages (like the P90/P10 ratio), or calculating the variance in wages. These numbers help us understand how big the wage gap is.

A recent graph shows that nearly 40% of workers in Turkey earn less than 1.05 times the minimum wage. This is the highest rate among the countries shown. In places like Slovenia and Bulgaria, this share is only around 13% to 15%. In most EU countries, it is even lower—under 10%. This means that a large number of Turkish workers are stuck at the bottom of the wage ladder. Their wages are very low, and they have few chances to move up. This is called wage compression. It means that wages are not fairly spread across workers. It also shows that the labor market in Turkey offers limited upward mobility. So, wage inequality in Turkey is not only about the rich earning more. It is

also about millions of workers being trapped in low-paying jobs.

Many factors cause wage inequality. Education is a big one. People with more education often earn more. Skills and work experience also matter. Technology has made this worsen as new machines and computers increase the need for skilled workers, but leave others behind. Another problem is Turkey's labor market because there are many informal jobs, and unions are weak. Globalization also plays a role. Companies move jobs to cheaper countries, putting pressure on local wages.

Wage inequality does not affect everyone equally. Some groups suffer more. Women, for example, still earn less than men for similar jobs. However, in our data, the gender wage premium appears to be close to zero. This means that, on average, women and men earn similar wages in Turkey according to recent statistics. Young people struggle to find good jobs. People in eastern and southeastern Turkey have fewer opportunities. These groups face long-term disadvantages, making it hard to escape poverty.

Wage inequality causes many problems. First, it slows down the economy. People with low wages cannot spend much, which reduces demand. Second, it makes it harder for people to move up in life. If you are born poor, you stay poor. Third, it creates anger and tension in society. People feel things are unfair. This weakens trust in the government and democracy.

Many economists have studied inequality. Thomas Piketty says that capital and wealth grow faster than wages, so rich people get richer. Anthony Atkinson believes we need better education and fair taxes. Raj Chetty shows that inequality makes it hard for people to improve their lives.

There are also some wrong ideas about inequality. Some say, "If you work hard, you will earn more." But this is not always true. Opportunities are not equal for everyone. Others believe that inequality makes people work harder. But in many cases, it does the opposite. When people feel they cannot win, they stop trying. Some think inequality is natural in a market economy. But policies and laws shape the market. Governments can reduce or increase inequality with their choices.

It is also important to know that wage inequality does not tell us everything. A country can be rich and still have high inequality, like the United States. On the other hand, a country with low inequality might still face problems. Wage inequality is just one part of the story, but it is an important one.

2 Data and Methodology

To construct the ground on which the analyses depend, what the data are, how to classify and make them usable for the estimation, and how to deal with them will be searched. For the analysis, two different official data will be used: the Household Budget Survey (HBS) and the Survey of Income and Living Conditions (SILC), which are valid data for the methodology that will be used. The same methodological framework with Review of Economic Dynamics (RED) guidelines for inequality studies will be applied. Moreover, the to-be-calculated inequality measurements, inequality premiums, and annual earning estimations also comply with RED guidelines.

To be able to analyze the evolution of wage inequality in Turkey since 2002, HBS and SILC will be used extensively. Both data come from the Turkish Statistical Institute, which is the official institute. HBS and SILC are annual surveys that try to get data on income, consumption, and earnings from households and are constructed to be representative of Turkey. For the estimation for 2022, HBS in 2022 and SILC in 2023 are used (the publication year consists of the previous year's data in SILC). HBS data includes disposable income and consumption subcomponents at the household level and labor supply and income sources at the individual level, such as labor earnings, asset income, self-employment, and government/private transfers. The focus is on labor earnings and self-employment. HBS data surveys 39,189 persons in 2022 and is nationally representative. SILC has been conducted since 2006. It covers detailed income sources for all household members. Labor earnings and self-employment are focused on here too. SILC data surveys 60,223 persons in 2022 and is nationally representative. Both HBS and SILC comply with RED guidelines for inequality studies (Tamkoç and Torul (2020)).

For the estimation, all monetary variables are converted to real terms using CPI 2002 as a base. Because wage is concerned, working-age individuals(25-60 years) are used, and individuals whose annual earnings are below 2439 TL are excluded due to the fact that 2439 TL in 2022 is approximately equal to 50% of Turkey's minimum wage at the time, which is done to be able to focus on the wage inequality of earnings of people who work 'regularly.'

To be compatible with RED guidelines, annual earnings are calculated as follows:

$$a_{e,i,t} = nw_{i,t} + rw_{i,t} + p_{i,t} + b_{i,t} + \alpha_{\text{TUR}}(nse_{i,t} + rse_{i,t} + ag_{i,t})$$

where $a_{e,i,t}$ denotes annual earnings, $nw_{i,t}$ and $rw_{i,t}$ denote annual cash and other real payments, $p_{i,t}$ and $b_{i,t}$ denote annual premiums and bonuses received from employers, α_{TUR} denotes the share of labor income in Turkey's national income, $nse_{i,t}$ and $rse_{i,t}$ denote cash and other real incomes from self-employment, and $ag_{i,t}$ denotes net agricultural income.

To make the estimation richer and to make comparisons between subgroups in the population, inequality premiums will be used. To that end, hourly is consulted to. Hourly wage is calculated as earnings divided by hours worked. Gender wage premium refers to the portion of wage differences between men and women that cannot be explained by observable characteristics. Education premium represents the additional earnings workers receive for each additional year of schooling or higher degree attainment. The premium reflects both productivity gains from education and credential effects. Experience premium captures the wage increases associated with accumulated work experience. The premium reflects both skill acquisition and signaling value of sustained employment. Hourly wage and the inequality premiums are calculated as follows:

Hourly Wage Calculation:

$$w_{i,t} = \frac{ae_{i,t}}{ah_{i,t}}$$

where $ae_{i,t}$ is Annual earnings, $ah_{i,t}$ is Annual hours worked

Wage Premiums:

Premium Type	**Definition**	
Gender Premium	$\overline{w_{m,t}}/\overline{w_{f,t}}$	(Male/Female avg. wage)
College Premium	$\overline{w_{c,t}}/\overline{w_{h,t}}$	(College/HS grad males)
High School Premium	$\overline{w_{h,t}}/\overline{w_{s,t}}$	(HS/Secondary school males)
Experience Premium	$\overline{w_{o,t}}/\overline{w_{y,t}}$	(Age 45-55/25-35 males)

where $\overline{w_{x,t}}$ denotes average wage for group x in year t

Gini coefficient, variance of residuals, percentile ratios(P90/P50, P50/P10) and variance of log wage are used to estimate the wage inequality in the country. These differ in a way that each of them assigns different weights to different parts of the population. The variance of residuals measures wage inequality that persists after accounting for observable worker characteristics (e.g., education, experience, industry) in a regression model. It captures disparities stemming from unobserved factors such as discrimination, unmeasured skills, labor market imperfections, or pure randomness. The gini coefficient quantifies overall wage inequality across the entire income distribution, ranging from 0 (perfect equality) to 1 (one person earns all income). It is derived from the Lorenz curve, which plots cumulative wage shares against the population.It is equal to the area that is between line of perfect inequality and Lorenz line multiplied by two. Percentile ratios dissect inequality by comparing specific points in the wage distribution. P90/P50 (top 10% vs. median) and P50/P10 (median vs. bottom 10%) isolate upper- and lower-tail inequality, respectively. This metric measures total wage dispersion using log-transformed wages, excludes the influence of extreme values and ensures proportional interpretation. Variance of residuals and variance of log wage calculated as follows:

- **Variance of Residuals:**

$$\log(y_{i,t}) = c_t + \beta_1 D^s_{i,t} + \beta_2 D^e_{i,t} + \beta_3 D^f_{i,t} + \beta_4 D^a_{i,t} + \beta_5 D^{ce}_{i,t} + \epsilon_{i,t}$$

  Then, calculate $\hat{\epsilon}_{i,t}$, residual error term.

- **Control Variables:**

Symbol	Description
D^s	Gender dummy
D^e	11 education dummies
D^f	Family composition dummies
D^a	7 age category dummies
D^{ce}	4 couple education dummies

- **Variance of Log Wage:**

$$\mathrm{Var}(\log W) = \frac{1}{N}\sum_{i=1}^{N}(\log w_i - \overline{\log w})^2$$

3 Results

3.1 Findings and Comments

In this section, we are going to look over the general condition of wage inequality and the social factors affecting the wage inequality in Turkey. First of all, to make the analysis academically coherent with the previous research. We followed the RED guidelines to assess the wage earners in both the SILC and HBS data. We included the individuals between 25–60 years old. We excluded the individuals who didn't report how many hours worked. Finally, excluded individuals whose annual earnings are below 2,439 Turkish Liras. (i.e. average half of Turkey's monthly minimum wage in 2022**?**).

To begin with, we made a descriptive statistics table to understand the main metrics from both datasets. The results can be seen from the table below. The most surprising factor about them is the huge difference in mean and median wages between both datasets. Moreover, while the mean wages are not so indifferent between gender groups, the max values are male for both of the datasets.

	Min	**Max**	**Mean**	**Median**	**Std. Dev.**
SILC - Overall	1.3846	1079	42.7545	31.7307	44.1467
SILC - Male	1.3846	1079	43.2685	31.7692	46.4847
SILC - Female	1.4690	807	41.3580	31.1965	37.0245
HBS - Overall	5.8080	633	136.8628	115.9861	99.4841
HBS - Male	5.8500	633	137.4669	116.6666	98.3245
HBS - Female	5.8080	625	135.1767	112.5000	102.6521

All values are TL — 1% Threshold is removed from HBS to remove outliers

We did a comprehensive analysis capturing the main metrics to analyze the wage inequality for both of the datasets. We checked the Gini coefficient, variance of log wages, variance of residuals, education, gender and experience premiums, P90/P50 and finally P50/P10 values. We made this analysis for both of the datasets and compared the results with the findings of Orhan Torul and Mehmet Nazım Tamkoç (2020). Our results mostly correlate with the results, with slight differences.

	Gini	**Var(Log Wages)**	**Edu. Premium**	**Gender Premium**	**Exp. Premium**	**P90/P50**	**P50/P10**	**Var(Residuals)**
SILC	0.4120	0.6304	1.6463	1.0461	1.0509	2.5454	2.8448	0.4605
HBS	0.4304	0.6983	1.5623	1.0050	0.9175	2.3535	2.4871	0.5058

Table 1: Our estimations for 2022

	Gini	Var(Log Wages)	Edu. Premium	Gender Premium	Exp. Premium	P90/P50	P50/P10	Var(Residuals)
SILC	0.4241	0.6580	1.5441	1.0265	1.0037	2.5013	2.7165	0.4898
HBS	0.4173	0.6480	1.5884	1.0095	0.9046	2.3413	2.8752	0.5902

Table 2: Torul's estimations for 2022

Furthermore, we deepened our analysis by looking at the regional average hourly wage and regional Gini coefficients. We used the SILC data for the analyses. We did this analysis by looking at the SR-1 regional levels, because the SILC data didn't have data for all SR-2 regions. The excluded regions were mostly those damaged by the 2023 Kahramanmaraş Earthquakes, so we decided to analyze them within the SR-1 regions. First, we look at the regional average hourly wages. As can be seen from the table below, the region with the lowest hourly wage is Southeast Anatolia. These results also correlate with the SEGE 2017 **?** results, which show that the least developed regions in Turkey are Southeast and East Anatolia.

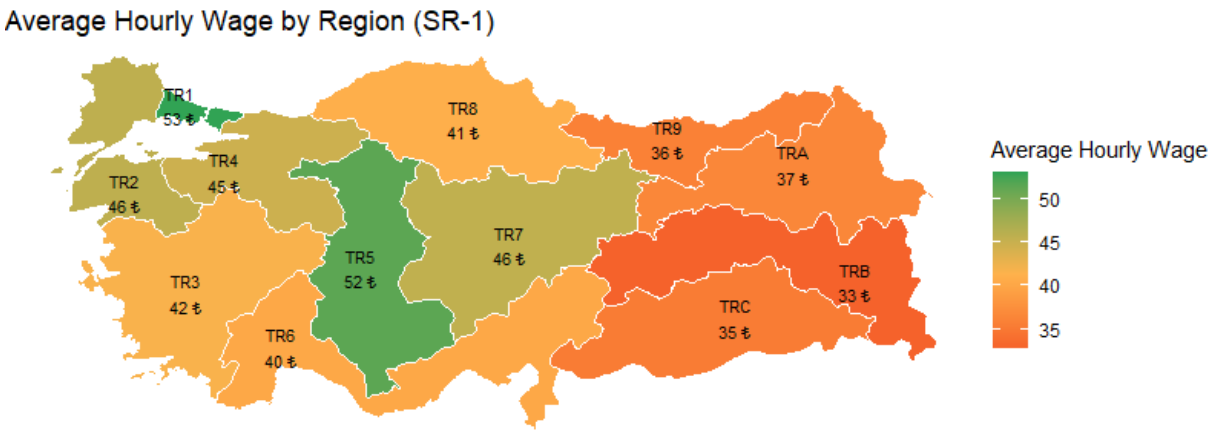


Figure 1: Regional Analysis of Average Hourly Wage

Also, we looked at the reginal gini coefficients. The figure below shows that the wage inequality in the boundaries of SR-1 regions. It can be seen that the worst part is different in regional gini. The least equal part is the Northeast part of the anatolia, especially East Black Sea region.

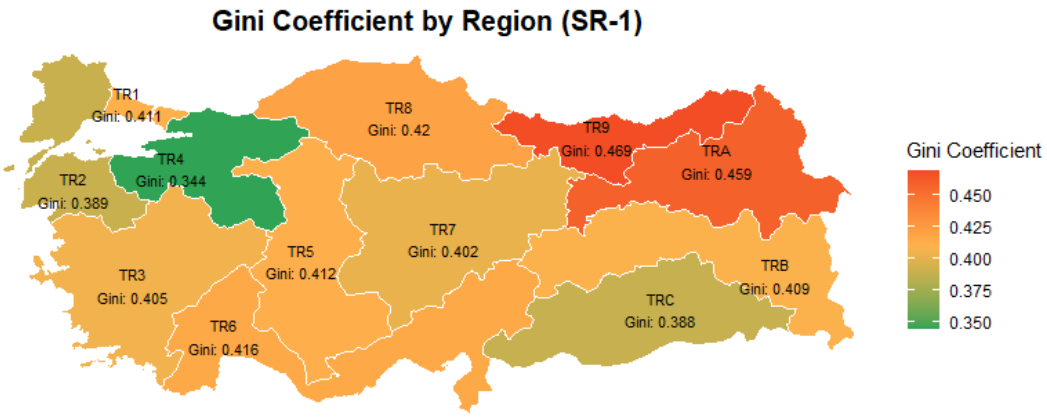


Figure 2: Regional Analysis of Gini Coefficient

To understand the wage inequality between age groups, we also checked the Gini coefficient of the age groups. We did this analysis for both of the datasets and divided the age groups as follows: 25–35, 35–45, and 45–60. While there are some differences in Gini coefficients between these two datasets, their pattern is the same. In both of the datasets, the Gini coefficient is lower in the younger group and increases as age increases. This can be explained by common sense: as people start to work, they tend to earn more equal amounts due to a lack of experience. But over time, as people gain experience and begin to specialize in their jobs, wage inequality grows.

Age Group	SILC	HBS
25–35	0.3649	0.3575
35–45	0.4083	0.3648
45–60	0.4641	0.4142

Furthermore, we wanted to analyze the informality in Turkey, which seems to be an important factor affecting wage inequality. Using the SILC dataset, the summary statistics for employment type is created, which can be seen from the table below. We used the registration status to Social Security Administration as an indicator of formal employment. The informal employment rate among wage earners is approximately 22%, which has declined to this level over the past decades according to previous studies in

Turkey, such as Bakis and Polat (2021). The more interesting finding is that informal workers earn almost half of formal workers on average, whereas the maximum values are similar.

Group	**Mean**	**Median**	**Min**	**Max**	**Share**
Formal	48.20	36.54	1.38	1079.06	77.9%
Informal	23.59	16.48	1.44	807.69	22.1%

Finally, we decided to implement a regression analysis to understand the reasons behind the wage inequality. We used a simple but effective model, looking at the relationship between years of schooling, being female, age and age^2, and finally used regional factors from the SILC data. Unfortunately, the HBS data didn't have regional classification, so we did the regression without regional factors for HBS.

$$\begin{aligned}\log(\text{hourly wage}_i) = \beta_0 + \beta_1 \cdot \text{schooling}_i + \beta_2 \cdot \text{female}_i \\ + \beta_3 \cdot \text{age}_i + \beta_4 \cdot \text{age}_i^2 + \sum_{r=1}^{R} \delta_r \cdot \text{Region}_{ir} + \varepsilon_i\end{aligned}$$

The methodology can be seen from the equation above. We used the $\log(\text{hourly wage}_i)$ as the dependent variable. We used β_0 as the intercept, β_1 as the coefficient for years of schooling, β_2 as being female, β_3 as age, β_4 as age^2, and finally δ_r as regional factors. All regions are compared against the reference category, TR1 (Istanbul Region). The results are in the table below.

As expected, years of schooling has a strong and statistically significant positive effect on wages in both datasets, indicating higher returns to education. Similarly, being female is associated with lower wages, although the magnitude is relatively small. The relationship between age and wages follows a concave pattern, as shown by the negative coefficient on age and the positive coefficient on age squared—suggesting that wages increase with age but at a decreasing rate.

In terms of regional effects, nearly all coefficients for the regional dummies are negative and significant, indicating that workers in other regions earn significantly less than those in Istanbul. The regional wage gap is especially pronounced in less developed regions such as TRC (Southeast Anatolia) and TRA (Northeast Anatolia), which show the largest negative coefficients. This finding aligns with the observed spatial inequality in Turkey and reinforces the wage disparity between metropolitan and peripheral regions.

	Dependent variable:	
	Log(Hourly Wage)	
	HBS 2022	SILC 2023
	(1)	(2)
Years of Schooling	1.8148***	1.2368***
	(0.1522)	(0.0917)
Female	0.0716***	0.0856***
	(0.0016)	(0.0011)
Age	−0.1821***	−0.1850***
	(0.0171)	(0.0108)
Age Squared	0.1134***	0.0806***
	(0.0072)	(0.0045)
age2	−0.0013***	−0.0009***
	(0.0001)	(0.0001)
TR2: Batı Marmara		−0.1318***
		(0.0239)
TR3: Ege		−0.2005***
		(0.0193)
TR4: Doğu Marmara		−0.0857***
		(0.0207)
TR5: Batı Anadolu		−0.1009***
		(0.0210)
TR6: Akdeniz		−0.2597***
		(0.0215)
TR7: Orta Anadolu		−0.1630***
		(0.0237)
TR8: Batı Karadeniz		−0.2905***
		(0.0211)
TR9: Doğu Karadeniz		−0.4784***
		(0.0288)
TRA: Kuzeydoğu Anadolu		−0.4137***
		(0.0238)
TRB: Ortadoğu Anadolu		−0.4226***
		(0.0233)
TRC: Güneydoğu Anadolu		−0.2613***
		(0.0207)
Observations	8,979	20,002
R^2	0.2265	0.2913
Adjusted R^2	0.2261	0.2908
Residual Std. Error	0.7108 (df = 8974)	0.6687 (df = 19986)
F Statistic	656.8368*** (df = 4; 8974)	547.6401*** (df = 15; 19986)

*p<0.1; **p<0.05; ***p<0.01. Standard errors in parentheses.

Table 3: Regression Results

3.2 Comparison with the World

Recent empirical analysis highlights that Turkey exhibits higher levels of wage inequality than most developed countries.

Table 4: Wage Inequality and Wage Premia across Countries

Country	Level in year 2000				Change				
	Var. log w	College premium	Exp. premium	Gender premium	College premium	Exp. premium	Gender premium	Var. log w	Period
Canada	0.45	1.80	1.32	1.33	0.22	0.31	-0.11	0.17	1978-2006
Germany	0.27	1.38	1.27	1.28	-0.08	0.22	-0.15	0.05	1983-2003
Italy	0.17	1.51	1.34	1.03	-0.08	0.11	-0.05	0.03	1987-2006
Mexico	0.62	1.88	1.23	1.21	0.40	0.22	-0.06	0.04	1989-2002
Russia	0.77*	1.50	1.05*	1.49	-0.06	0.05*	-0.07	-0.13*	1998-2005
Spain[a]	0.23	1.48	1.43	1.16	-0.33	0.07	-0.21	-0.18	1985-1996
Sweden[b]	0.18	1.61	1.20	1.22	0.14	-0.02	-0.05	-0.09	1990-2001
UK	0.33	1.62*	1.25*	1.32	0.12*	0.20*	-0.21	0.10	1978-2005
USA	0.44*	1.80*	1.38*	1.36	0.40*	0.28*	-0.25*	0.21*	1980-2006
Average	**0.38**	**1.62**	**1.27**	**1.27**	**0.11**	**0.17**	**-0.10**	**0.04**	

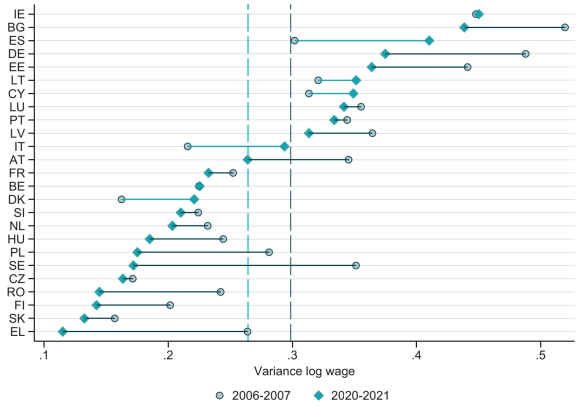
A * indicates the statistic is from data on males only. Wage premia and wage dispersion for women are typically smaller.
(a) Data on changes refer to after-tax annual earnings
(b) Data on levels is for 1992

Source: Krueger et al. (2010)

Using RED-compliant methods, we found that Turkey's variance of log wages is 0.6304 (SILC) and 0.6983 (HBS), which has declined to these levels over the past twenty years, said by Tamkoç and Torul (2020). These estimations exceed those observed in many countries for the level in year 2000, despite a lower gender wage premium (SILC: 1.0461; HBS: 1.0050), which may result from the low female labor force participation (under 27%). Thus, even though we have observed a significant decline in wage inequality, it is still higher than that of developed countries' wage inequality levels in year 2000.

In addition, as Zwysen (2024) notes, Europe has experienced a general decline in wage inequality, similar to Türkiye. Thus, variance log wage of European countries is still lower than that of Türkiye though there might be methodological differences for the estimations of Zwysen.

Figure 3: Change of Variance Log Wage in Europe



Source: Zwysen (2024)

Internationally, educational and experience-based wage premiums in Turkey remain significant. Our data indicate an education premium of 1.6463 (SILC) and 1.5623 (HBS), and an experience premium of 1.0509 (SILC) and 0.9175 (HBS), highlighting persistent skill-based earnings gaps. Percentile ratios such as P90/P50 (2.5454 in SILC) and P50/P10 (2.8448 in SILC) further confirm this inequality pattern.

In conclusion, Turkey's wage inequality is not only quantitatively high but also qualitatively distinct. The relatively small gender premium masks deeper structural issues, including informality, regional disparities, and educational returns. These findings call for policy attention focusing on education quality, formalization of employment, and inclusive labor market reforms.

4 Review of the Literature and Conclusions

In the final section, we reviewed the literature to understand the trend in wage inequality in Turkey and the impact of the factors.

Firstly, it is argued, in the literature, that real minimum wage growths are very impactful on wage inequality, both in Turkey and Brazil.

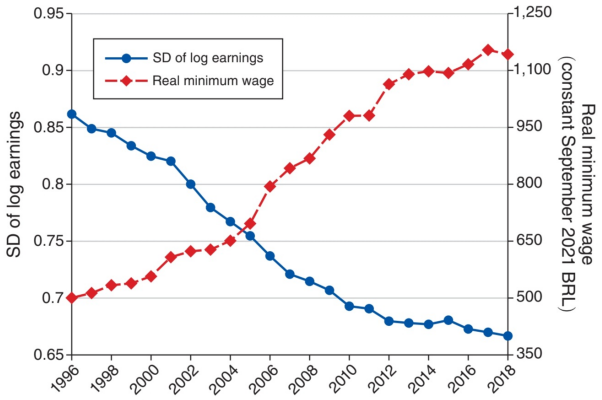


FIGURE 2. EVOLUTION OF WAGE INEQUALITY AND THE REAL MINIMUM WAGE

Notes: Statistics are for males of age 18–54. Real minimum wage is the annual mean of the monthly time series. The correlation between the two time series is −0.973. SD is standard deviation.

Sources: RAIS (1996–2018); IPEA (1996–2018)

Source: Engbom and Moser (2022)

They assert that real minimum wage growth accounts for almost 45% of the decline in wage inequality in Brazil. Similarly, Tamkoç and Torul (2020) mention that increasing real minimum wage is the main reason for decreasing wage inequality over the past twenty years because rate of minimum wage earners is relatively high[1] and the average real wages do not increase as much

[1]Approximately 40% according to EuroStat

as the real minimum wage, thereby bottom of the distribution converges to the mean, which decreases the wage inequality. The increasing real minimum wage can be seen from the table below.

Year	Annual CPI (2002=100%)	Monthly Minimum Wage Net (Real)
2002	100.00%	173.91
2003	121.60%	185.85
2004	132.10%	235.17
2005	142.90%	245.03
2006	156.60%	242.95
2007	170.30%	241.39
2008	188.10%	261.78
2009	199.80%	268.67
2010	216.90%	271.02
2011	231.00%	278.98
2012	251.50%	286.47
2013	270.36%	291.59
2014	294.30%	295.11
2015	316.85%	307.66
2016	341.15%	381.35
2017	379.57%	369.91
2018	441.57%	363.05
2019	508.58%	397.36
2020	571.03%	407.11
2021	682.93%	413.79
2022	1176.75%	414.44
2023	1805.59%	551.32
2024	2889.64%	588.38
2025	4024.76%	549.20

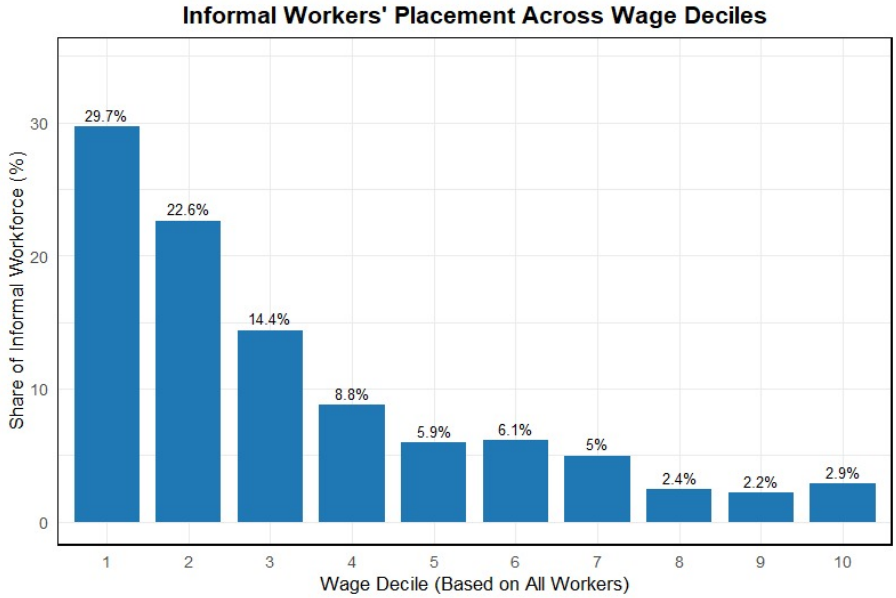
Table 5: Annual CPI and Real Net Minimum Wage (2002–2025).
Source: TurkStat

In addition, GDP growth is seemed to be negatively correlated with wage inequality in Turkey (i.e. wage inequality is counter-cyclical) according to Tamkoç and Torul (2020).



Source: Torul's estimations for Gini coefficient and TurkStat for GDP growth

Finally, they argue the role of informality in wage distribution. It is said that declining informality rate has an effect on decreasing wage inequality to some extent. The main reason is



considered to be the fact that informal workers are at the bottom of the wage distribution, which can be seen from the table above. Also, it should be recalled that average wage of informal workers is almost half of the average wage of formal workers. Therefore, as informality rate decreases[2], wage inequality mechanically decreases as well.

[2]Demonstrated by Bakis and Polat(2021)

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